THESIS DISCUSSION · PROF. PRENDINGER

M2 Target Comparison

Triple-barrier labels · four backends · 20 crypto assets

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SETUP Daily candles · frozen Kronos · cash-MTM · day-7 exits

BASELINE M1 UP Sharpe: −2.0372

UPDATED 29 October 2026 · 48 / 48 fits reviewed

Data Splits

Split	Start	End
Train	2024-06-30	2025-08-31
Validation	2025-09-08	2025-12-31
Test	2026-01-08	2026-07-18

Test dates are entry dates. Final day-7 exits: 2026-07-25.

VERIFIED AGAINST PABLO'S PAPER

M1 Baseline

Cash-MTM · day-7 exits · 0.2% fee · valued through 2026-07-25.

M1 side	Return	Sharpe	Max drawdown	Trades
UP	−43.31%	**−2.0372**	−48.75%	1045
DOWN	12.83%	0.7622	−27.84%	2580

48 / 48
M1 baselines match the paper

28
Weekly returns
199 daily bars

−40.01%
Buy-and-hold return

UP loses money; DOWN gains. All six targets share the same M1 Test trades.

roi_ext_mtm.csv reproduced: same M1 trades, engine and final exits. Annualized seven-day Sharpe ×√(365.25/7); 28 observations.

PAPER BACKTEST

MTM Valuation

1045 UP trades · all positions closed by 2026-07-25.

Equity calculation	Total return	Max drawdown	Sharpe
Cash · positions held at entry cost	−43.31%	−45.35%	−2.72
Cash-MTM · paper method	−43.31%	−48.75%	−2.04

Same final return

−43.31% after every trade closes.
Same trades and capital allocation.

Different path

MTM values open positions at each close.
It changes drawdown and Sharpe.

Entries end 2026-07-18 exits continue for 7 days. M1 requests 10% of free cash per trade; oversubscribed entries fill pro rata.

Pablo's Results

Kronos · 1d · TP labels · final exits 2026-07-25.

Strategy	UP return	UP Sharpe	DOWN return	DOWN Sharpe
M1 · all calls	−43.31%	**−2.04**	12.83%	0.76
Random Forest	−24.66%	**−1.20**	9.74%	1.14
AutoGluon	−34.26%	**−1.53**	9.90%	1.06
CTTS	−29.60%	**−1.37**	5.83%	0.52
TabPFN	−23.87%	**−1.18**	11.67%	1.21
TabICL	−33.91%	**−1.59**	14.80%	1.36

M1 matches exactly. Our M2 models are new fits; their predictions differ from these saved models.

roi_ext_mtm.csv; 150 replay checks across 10 Kronos daily cells. Same engine, fees and exit calendar for the following target comparison.

From Prices to Labels

Past prices
Frozen M1 proposes UP or DOWN.

→

Future prices
Apply exits and fees.

→

Meta-label z
1 if profitable; 0 otherwise.

Inspect labels

Compare exits and label changes on historical M1 calls. No retraining needed.

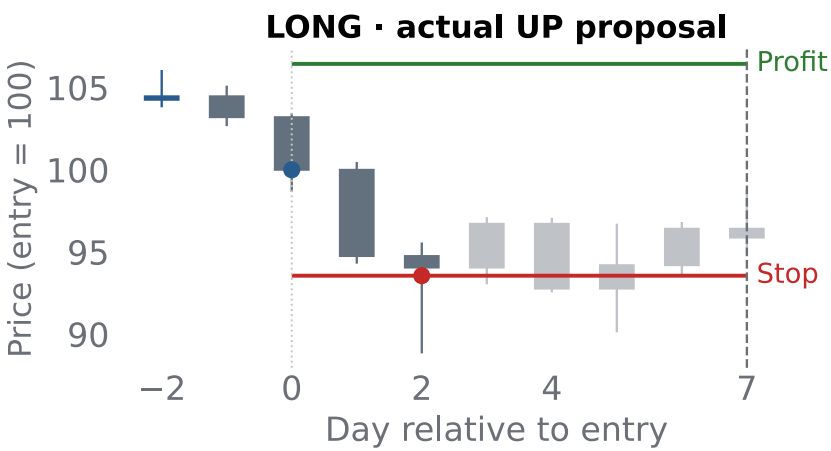
Test performance

Retrain M2 on each target, then compare. Keep M1 frozen.

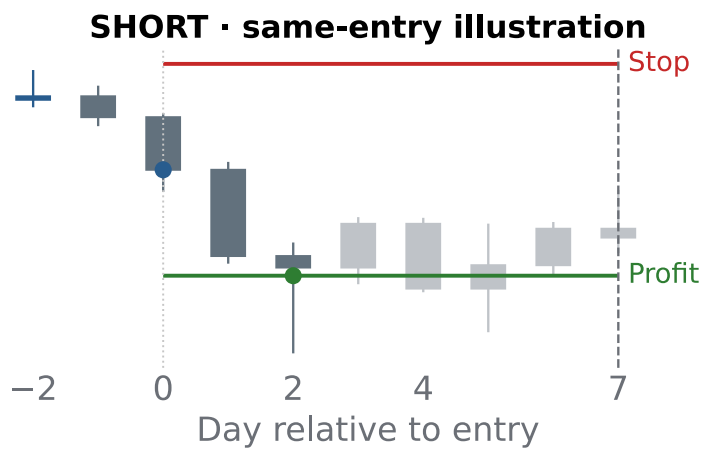
AFML §§3.4–3.7. Entry dates: 2024-07-01 – 2025-08-24. 7,593 proposals; 19/20 assets (TON unavailable).

Long and Short Barriers

BTC · 2024-07-03 · entry at day-0 close · symmetric k = 1 barriers.



Exit day 2 · net -6.6% · meta-label 0



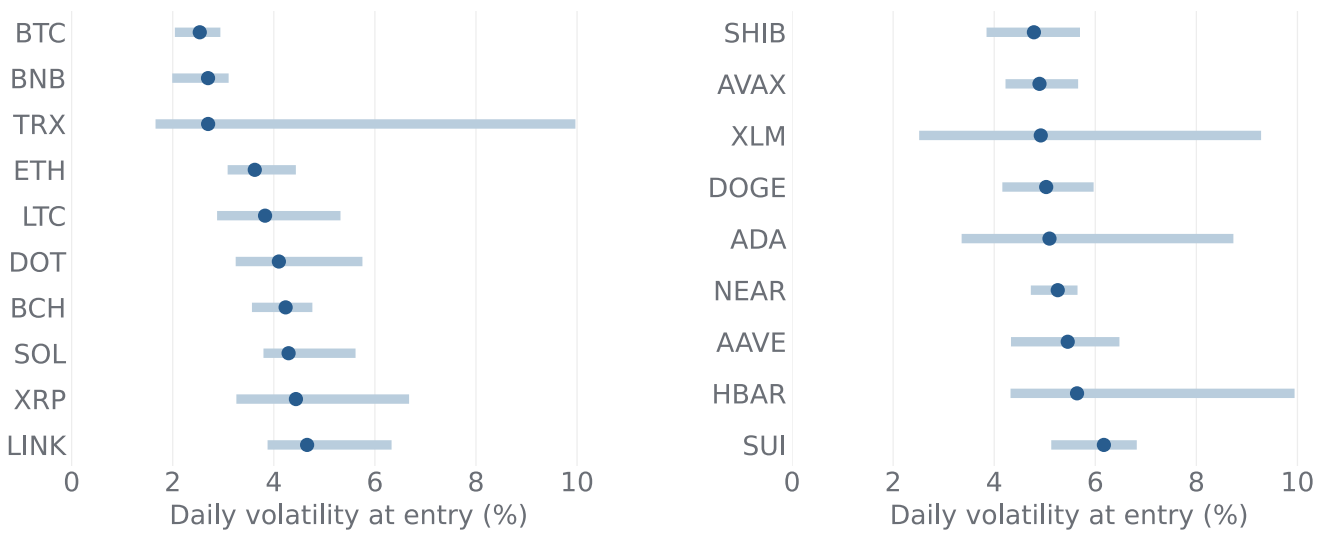
Exit day 2 · net +6.2% · meta-label 1

Long: profit above entry. **Short**: profit below entry. Stops are on the opposite side.

Training example selected for a stop; SHORT is counterfactual. Same volatility for both sides; 0.2% cost; assumed barrier fills.

Barrier Width by Asset

$$w_{a,t} = k\sigma_{a,t}\sqrt{7}$$

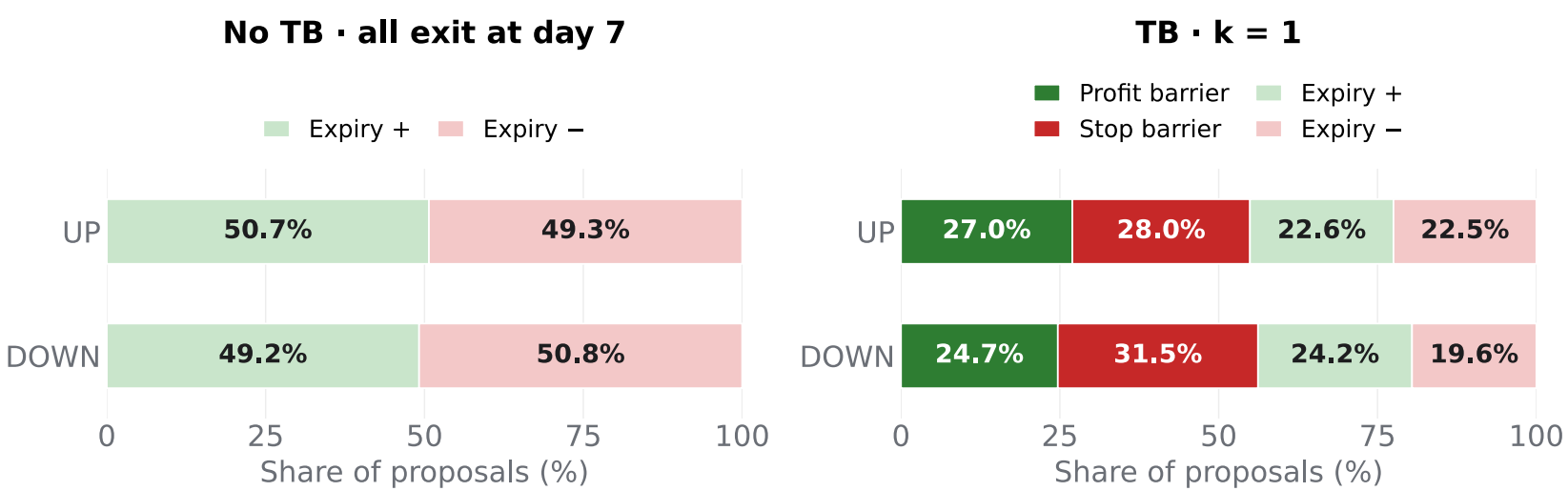


Same k, different widths: **BTC 6.7% · ETH 9.6% · SUI 16.3%** at k = 1.

Entry dates: 2024-07-01 – 2025-08-24. EWMA(100) at entry; width stays fixed per trade. Dots: median; lines: 10th–90th percentiles.

UP and DOWN Outcomes

1,921 UP proposals · 5,672 DOWN proposals · positive/negative after 0.2% fees.

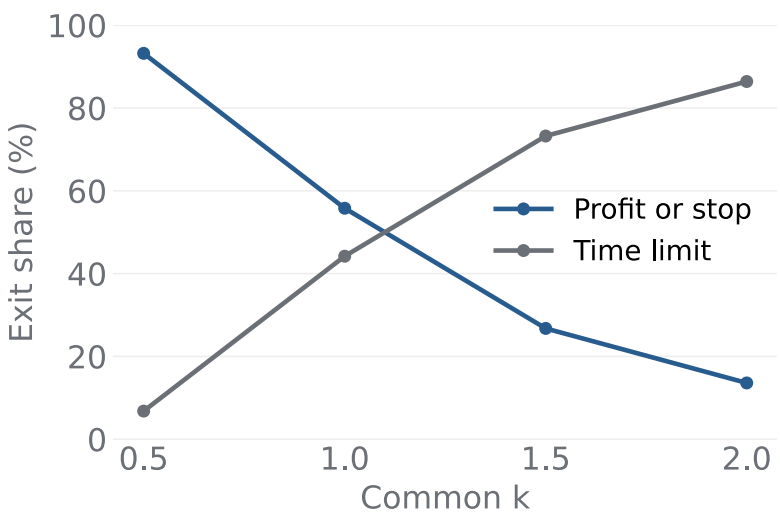


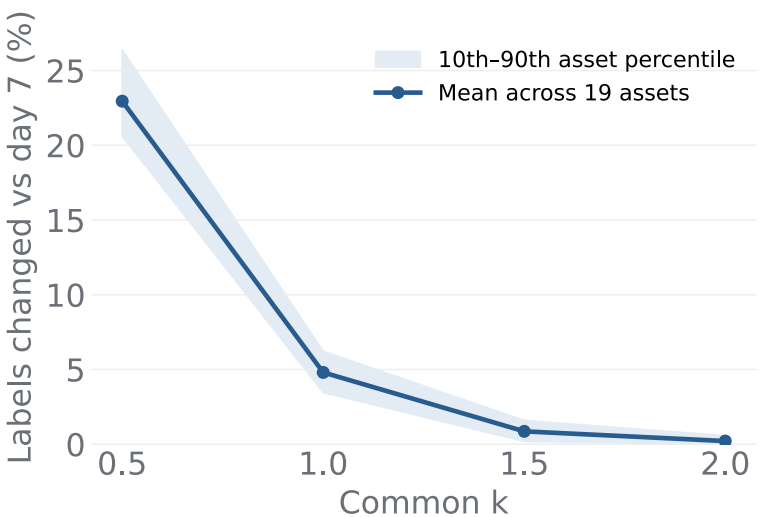
Labels change for **5.9% of UP** and **4.4% of DOWN** proposals.

Entry dates: 2024-07-01 – 2025-08-24. Each bar = 100% of that side's proposals. Expiry +/−: positive/negative net return. High/low touches; same-candle ties = stop.

Sensitivity to k

Small k: closer exits. Large k: more trades remain open until day 7.





At k = 1: **55.8% profit/stop exits**, yet only **4.8% of labels change**.

Entry dates: 2024-07-01 – 2025-08-24. Equal asset weights; shading: 10th–90th percentiles. Training diagnostics do not select an optimal k.

Choosing k

One shared k

- All assets and both directions.
- Test 0.5, 1, 1.5, 2.
- Fix features, horizon and fees.

Validate the choice

- Define exits and the objective first.
- Check return, drawdown and stability.
- Confirm on fresh data.

Volatility already adapts the width. Add more parameters only with a stable validation gain.

Proposed protocol: AFML ch. 7 and 11. The 2026 Test has already been inspected; label balance alone is not an objective.

TP and TB Labels

Frozen M1 proposes trades; TB builds labels for M2. TP checks profit at day 7 after fees.

Label 1

Positive net return at day 7.

Label 0

Zero or negative net return at day 7.

Column	M2 training target	Purpose
TP	Original day-7 label	Baseline
TB ref	TP label on TB-eligible samples	Control for the sample filter
TB k	Net outcome at profit / stop / expiry	Test a different label

TP = true-positive meta-labeling, not take profit. All backtests keep day-7 exits.

TP rule verified on 18,230 complete proposals. Labels use future outcomes; M2 inputs use only information available at entry.

Test Sharpe

Cash-MTM · entries 2026-01-08 – 2026-07-18 · final exit 2026-07-25.
M1 Sharpe: **UP −2.0372** · DOWN 0.7622. RF = Random Forest.

M2 fit	TP	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	−1.329	−1.581*	−1.131	−1.638*	−1.439	−1.379
AutoGluon · DOWN	−0.126	0.695	−0.217	0.372	0.735	0.414
Kumo · UP	−1.424	−1.533	−1.666	−1.881	−1.466	−1.558
Kumo · DOWN	0.517	0.404	−1.122	0.511	0.710	0.398
TabPFN · UP	−1.651	−1.682	−1.647*	−1.834	−1.605	−1.717
TabPFN · DOWN	0.789	0.743	0.825	1.161	0.679	0.635
RF · UP	−1.105	−0.929	−1.321	−0.991	−0.839	−0.975
RF · DOWN	0.624	0.780	0.026	0.279	0.789	0.558

All UP scores are negative. Highest observed DOWN: **TabPFN k = 1 (1.161)**.

Annualized cash-MTM Sharpe; 28 weekly returns. * = original threshold fell back to 0.5. One fit per cell; exploratory results.

SAME BACKEND AND DIRECTION

Change vs TP

Δ Sharpe = target − TP · positive values mean improvement.

M2 fit	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	−0.252	+0.199	−0.309	−0.110	−0.049
AutoGluon · DOWN	+0.821	−0.091	+0.498	+0.861	+0.540
Kumo · UP	−0.109	−0.243	−0.457	−0.043	−0.134
Kumo · DOWN	−0.113	−1.639	−0.006	+0.193	−0.119
TabPFN · UP	−0.031	+0.004	−0.183	+0.046	−0.067
TabPFN · DOWN	−0.047	+0.035	+0.372	−0.110	−0.154
RF · UP	+0.176	−0.216	+0.114	+0.266	+0.131
RF · DOWN	+0.155	−0.598	−0.345	+0.164	−0.066

DOWN: **AutoGluon k = 1.5 +0.861** · **TabPFN k = 1 +0.372** versus TP.

Includes the sample filter: TB removes **13 UP / 64 DOWN** early TON training proposals.

Cash-MTM through 2026-07-25 · 28 weekly returns. Same M1 trades, fees and sizing rule; fresh TP controls. Descriptive deltas.

SAME ELIGIBLE TRAINING SAMPLES

Change vs TB Reference

Δ Sharpe = TB-k − TB ref · compare labels on the same samples.

M2 fit	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	+0.451	−0.057	+0.142	+0.203
AutoGluon · DOWN	−0.912	−0.323	+0.040	−0.281
Kumo · UP	−0.134	−0.348	+0.067	−0.025
Kumo · DOWN	−1.526	+0.107	+0.306	−0.006
TabPFN · UP	+0.035	−0.152	+0.077	−0.035
TabPFN · DOWN	+0.082	+0.419	−0.064	−0.108
RF · UP	−0.392	−0.062	+0.090	−0.046
RF · DOWN	−0.753	−0.501	+0.009	−0.221

TabPFN DOWN · k = 1

Sharpe **1.161** · gain **+0.419**.
M1 DOWN: **0.762**.

RF UP · k = 1.5

Sharpe **−0.839** · gain **+0.090**.
Return **−18.15%**: still loses money.

Cash-MTM through 2026-07-25 · 28 weekly returns; one fit per cell. Gains vary by backend and direction; no optimal-k claim.